# Percentiles and Z-Scores

Name _____

You scored 85 on a test. Is that good? It depends on how everyone else did. Percentiles and z-scores tell where one value stands in a data set.

**Percentile**

The **percentile** of a value is the percent of the data that is **below** it.

$$\text{percentile} = \frac{\text{number of values below}}{\text{total number of values}} \times 100$$

**Example:** 20 students took a test. 15 students scored below 85.

$15 \div 20 = 0.75$, so a score of 85 is at the **75th percentile**. 85 is higher than 75% of the class.

**Z-score**

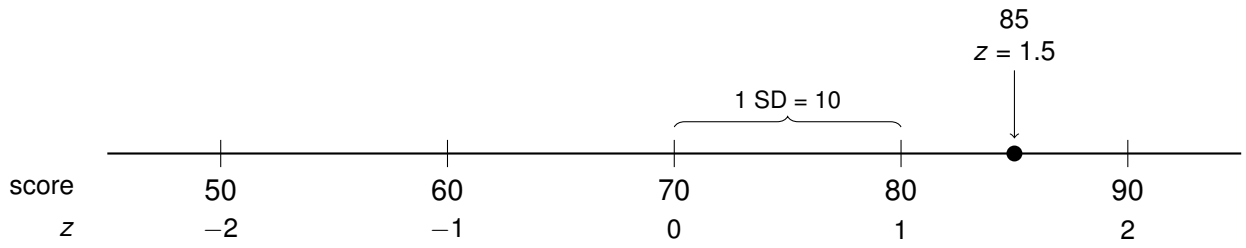
A **z-score** tells how many standard deviations a value is above or below the mean.

$$z = \frac{\text{value} - \text{mean}}{\text{SD}}$$

$z > 0$: above the mean. $z < 0$: below the mean. $z = 0$: equal to the mean.

**Example:** The test has a mean of 70 and an SD of 10. What is the z-score of 85?

$z = \frac{85 - 70}{10} = \frac{15}{10} = 1.5$ 85 is **1.5 standard deviations above** the mean.




**Comparing with z-scores.** Z-scores let you compare values from different data sets.

| | Score | Mean | SD | $z$ |
|---|---|---|---|---|
| Ana, math test | 85 | 70 | 10 | $\frac{85-70}{10} = 1.5$ |
| Ben, history test | 80 | 74 | 3 | $\frac{80-74}{3} = 2$ |

Ana's score is higher, but Ben did better **compared to his class**. His z-score is larger.

**Going backward:** value = mean + $z \times$ SD $z = -2$ on Ana's test: $70 + (-2)(10) = 50$

Work with a partner. Compare answers when you finish each problem.

1. Here are the quiz scores of 20 students, in order.

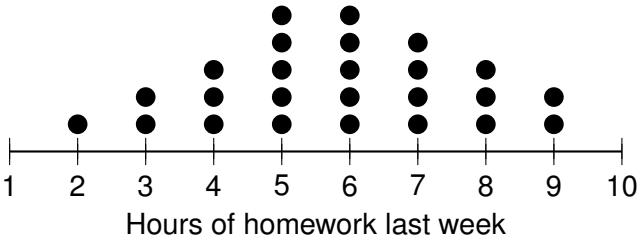
   52, 58, 61, 64, 66, 68, 70, 71, 73, 75, 76, 78, 80, 81, 83, 85, 88, 90, 94, 98

   (a) What percentile is a score of 83?

   (b) What percentile is a score of 61?

   (c) Which score is at the 50th percentile?

2. This dot plot shows how many hours 25 students spent on homework last week.




   (a) What percentile is 7 hours?

   (b) What percentile is 4 hours?

3. A class took a quiz. The mean is 20 points and the SD is 4 points. Find the $z$-score of each score. Then say what it means.

   (a) 26

   (b) 14

   (c) 20

   (d) 30

4. Lin took two tests.

| Test | Lin's score | Class mean | Class SD |
|---|---|---|---|
| Spanish | 88 | 80 | 4 |
| Science | 90 | 84 | 6 |

   (a) Find Lin's z-score on each test.

   (b) Lin's science score is higher. On which test did Lin do better compared to the class? Explain.

5. A test has a mean of 70 and an SD of 8.

   (a) What score has a z-score of 2?

   (b) What score has a z-score of $-0.5$?

6. In a 100-meter race, a **lower** time is better. The team's mean time is 14.0 seconds, with an SD of 0.5 seconds.

   (a) Maria ran 13.0 seconds. Find her z-score.

   (b) Is a negative z-score good or bad for Maria? Explain.

7. A student says, "A z-score of $-1$ means the value is 1 less than the mean." Is the student right? Explain.